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10 marks

```
enter the number of element:enter
the 5 element:largest element is
smallest element:1
```

CO1 - AT3 - Output Prediction

Back

QUESTIONS

- Q1** Largest and smallest element
10 marks
- Q2** Occurrence of an element
10 marks
- Q3** Searching
10 marks
- Q4** Array
1 mark

4/4 answered

Q2 Occurrence of an element

10 marks

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,x;
5     printf("enter size of array:");
6     scanf("%d",&n);
7     int arr[n];
8     printf("enter %d elements:",n);
9     for(int i=0;i<n;i++){
10         scanf("%d",&arr[i]);
11     }
12     printf("enter elements to find:");
13     scanf("%d",&x);
14     int first = -1, last = -1;
15     for(int i=0;i<n;i++){
16         if(arr[i]==x)
17         {
18             first=i;
19             break;
20         }
21     }
22     if (first!=-1){
23         for(int i=n-1;i>=0;i--){
24             if(arr[i]==x)
25             {
26                 last=i;
27             }
28         }
29     }
```

Input

7
2 5 3 5 7 5 8
5

Output

enter size of array:enter 7
element:enter elements to
find:first occurrence of 5 is at
index 1
last occurrence of 5 is at index
5

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DS PROGRAMMING

Hi Harshavardhan

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Q4

Array

1 mark

4/4 answered

```
6 scanf("%d",&n);
7 int arr[n];
8 printf("Enter %d elements:",n);
9 for(int i=0;i<n;i++){
10     scanf("%d",&arr[i]);
11 }
12 printf("Enter elements to find:");
13 scanf("%d",&x);
14 int first=-1,last=-1;
15 for(int i=0;i<n;i++){
16     if(arr[i]==x)
17     {
18         first=i;
19         break;
20     }
21 }
22 if (first!=-1){
23     for(int i=n-1;i>=0;i--){
24         if(arr[i]==x)
25         {
26             last=i;
27             break;
28         }
29     }
30 if(first!=-1)
31 {
32     printf("element %d not found in the array.\n",x);
33 }
34 }else{
35     printf("first occurrence of %d is at index %d\n",x,first);
36     printf("last occurrence of %d is at index %d.\n",x,last);
37 }
38 return 0;
39 }
```

Submitted

Output

```
enter size of array:enter 7
element:enter elements to
find:first occurrence of 5 is at
index 1
last occurrence of 5 is at index
5
```


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H Harshavardhan

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QUESTIONS

Q1

Largest and Smallest element

10 marks

✓

Q2

Occurrence of an element

10 marks

✓

Q3

Searching

10 marks

✓

Q4

Array

1 mark

✓

4/4 answered

Q4 Array

1 mark

Write a program for binary search

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,key;
5     printf("enter size of array:");
6     scanf("%d",&n);
7     int arr[n];
8     printf("enter %d sorted element:",n);
9     for(int i=0;i<n;i++)
10         scanf("%d",&arr[i]);
11     printf("enter element to search:");
12     scanf("%d",&key);
13     int low=0,high=n-1;
14     while (low<=high)
15     {
16         int mid=(low+high)/2;
17         if (arr[mid]== key){
18             printf("element found at index %d\n",mid);
19             return 0;
20         }else if (arr[mid]<key){
21             low=mid+1;
22         }else{
23             high=mid-1;
24         }
25     }
26     printf("element not found\n");
27     return 0;
28 }
```

Input

6
10 20 30 40 50 60
50

Output

enter size of array:enter 6 sorted
element:enter element to
search:element found at index 4